RESPONSIBLE AI

Bias & Fairness Visual Guide

Measuring and mitigating unfairness:
metrics, sources, and fixes across the pipeline.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Fairness Metrics

Different definitions — pick deliberately.

Demographic parity

equal approval rates across groups. Simple, often too strict.

Equalized odds

equal true/false positive rates across groups. Respects base rates.

Impossibility result: you cannot satisfy all fairness definitions at once when base rates differ.

Choose the metric with stakeholders BEFORE modeling. The metric choice is a values decision, not a math one.

STEP 2/5

Sources of Bias

Bias enters long before the model.

Historical bias

Training data reflects past discrimination (hiring, lending, policing).

Representation bias

Some groups are under-sampled; the model learns them poorly.

Measurement bias

Proxies mismeasure: 'arrests' is not 'crime', 'clicks' is not 'satisfaction'.

Deployment bias

A fair model used in an unfair workflow still produces unfair outcomes.

STEP 3/5

Pre-processing Mitigation

Fix the data before training.

- Reweighting: upweight underrepresented groups so the loss can't ignore them.
- Resampling: over/undersample to balance group representation in training.
- Disparate-impact remover: repair features to reduce group correlation while keeping rank order.
- Audit labels: have diverse reviewers check whether 'ground truth' encodes past bias.

Pre-processing is model-agnostic — one fix benefits every model trained on the data.

STEP 4/5

In-processing Mitigation

Build fairness into training.

Fairness constraints

optimize accuracy subject to a fairness bound. Explicit tradeoff.

Adversarial debiasing

a second network tries to predict the protected attribute — penalize it.

Removing the protected column is NOT enough — proxies (zip code, name) carry the signal. Measure, don't assume.

STEP 5/5

Post-processing and Auditing

Fix decisions and keep watching.

- Threshold tuning: set per-group decision thresholds to equalize the chosen metric.
- Reject option: route low-confidence, high-disparity cases to human review.
- Continuous auditing: recompute fairness metrics on live decisions — drift breaks fairness too.
- Document everything: model cards should state metrics, limitations, and intended use.

STEP ✓/✓

Key Takeaways

Bias & Fairness — Remember

- Pick a fairness metric deliberately — no model satisfies all definitions at once.
- Bias sources: historical, representation, measurement, and deployment.
- Mitigate at three stages: pre-process the data, constrain training, adjust decisions.
- Dropping the protected attribute doesn't work — proxies leak it.
- Audit continuously and document in model cards; fairness decays with drift.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev